

# LinkBuilder

## Outreach automation CRM

A link-building team's whole pipeline, run by one operator: finding donor sites, outreach, AI-drafted replies, payment requests and placement monitoring. Every AI draft passes versioned contracts before it can be sent.

**11,000+**

backend tests, full run green

**69**

contract rules on every AI draft

**16,000+**

placed links tracked

**20+**

background workers in one stack

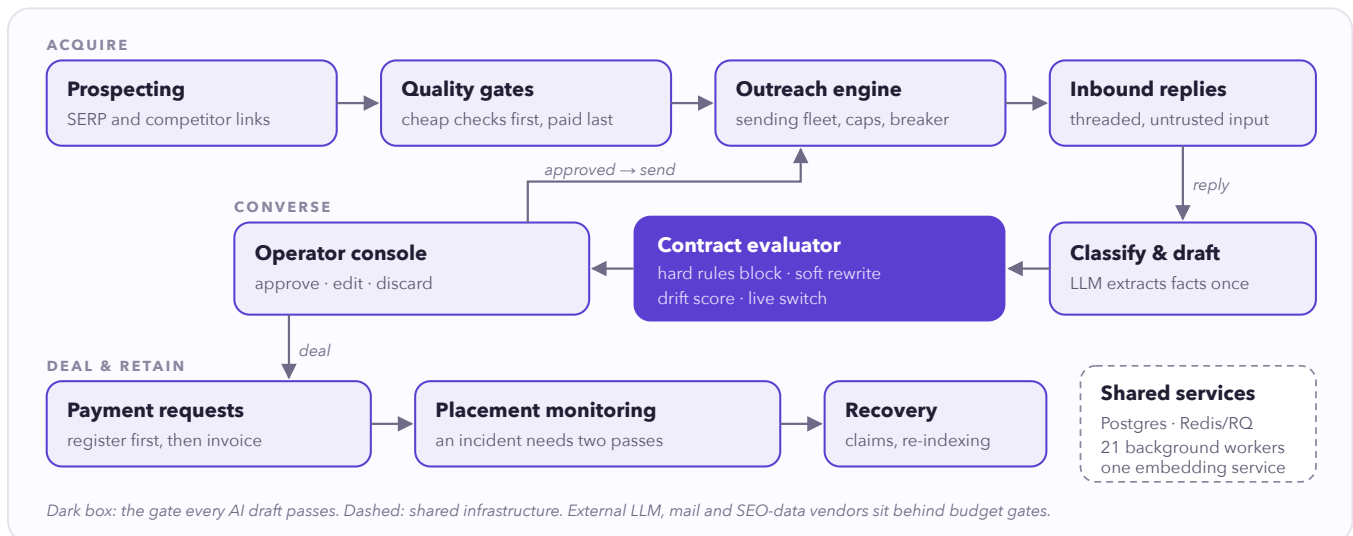
### THE PROBLEM

Link building is a long chain of paid, stateful steps, and failures look like successes: a drained data vendor once let a job report "600/600, 0 errors" after about 120 keywords. An LLM writing to strangers about money must obey rules that can be enforced, audited and switched off live.

### WHAT IT DOES

- Mines competitors' backlinks and gates every site, the paid check last
- Sends from a fleet of domains with caps and a bounce breaker
- Classifies each reply, drafts an answer, checks it against contracts
- Registers the payment request, then keeps verifying the placement

### ARCHITECTURE



### HOW IT FLOWS

- 1 Acquire** – discover sites, gate them, enrich contacts, launch a campaign.
- 2 Converse** – reply → classify → draft → contract check → a human approves → send.
- 3 Deal & retain** – payment runs register-first; an incident opens only after a confirming pass.
- 4 Control** – vendor balances raise alerts; a live switch sets checks to full, warn-only or off.

### WORTH KNOWING

#### 63 of 69 rules are plain code.

After seven rounds of prompt tuning, the LLM became a fact extractor and deterministic code became the judge.

#### Safe is not the same as useful.

The safety judge passed 109 of 109 shadow drafts; the operator sent only a quarter unchanged. Auto-send waits for replay data.

#### "Dead" is checked twice.

159 of 865 "dead" placements were alive – donors' firewalls were blocking the checker, not the links.

DECISIONS THAT SHAPE IT

**Correctness in the database, not the scheduler**

01

**Because** one hung send tick once stopped mail for four hours. A send is now claimed atomically and its message ID stored before the provider is called; a stuck send is re-queued only if the provider provably never took it.

**Cost:** hand-built retries and reapers.

**Contracts as versioned specs, not prompt rules**

02

**Because** the rules had drifted between a generator prompt and a validator prompt. A violation now forces a rewrite; if the last attempt still breaks a hard rule, the draft is blocked and escalated.

**Cost:** 69 invariants and 11 caps to maintain.

**Code judges, the LLM extracts**

03

**Because** LLM-as-judge kept its false positives through seven tuning rounds. The model now extracts thread facts once; code rules on them, and tests replay exactly.

**Cost:** one structured LLM call per thread.

**Autonomy earned by measurement**

04

**Because** passing the safety judge is not the same as being worth sending. Auto-send stays off until replay shows 70-80% of first replies sent unchanged.

**Cost:** operator time on every reply until then.

**Register first, then the side effect**

05

**Because** a failed invoice upload once rolled back a registration, and a second request went in for the same money. Registration now commits first, and a test pins the order.

**Cost:** a visible "registered, no invoice" state to retry.

**Fail loud on paid APIs**

06

**Because** a silent partial result costs more than a refused run. A drained vendor now stops the job with a reason, and runs refuse to start without balance.

**Cost:** runs refused when a balance is low.

HOW IT IS ENGINEERED

**Design doc**

62 of them, one per feature

→

**ADR**

with revisit criteria

→

**Build**

AI coding agents under my lead

→

**Gates**

16 on commit, 11 ratchets

→

**Verification**

16 reports and audits

→

**Handover**

the running system, not the design

**11,000+**

backend tests, full run green

**705**

frontend tests

**1,098 / 1,953**

commits co-written with AI agents, under the gates

**84**

adversarial prompt-injection cases

**Inbound mail is untrusted input.** Five layers stand between a webmaster’s email and the model: a sanitizer with 61 signatures, untrusted-content wrapping, a fail-closed contract gate, output guards against a redirected payment, and an adversarial suite. Attack payloads are logged only as hashes.

**WHERE IT TRANSFERS**

The pattern is not about SEO. Any business that lets an LLM write to customers – sales, support, collections, procurement – needs the same layers: the model proposes, versioned contracts check deterministically, a human approves until replay data says otherwise, and an admin can stop everything without a deploy. Scale comes from the boring parts: a stateless API, queue workers that scale independently, correctness held by atomic database claims, and budget caps on every paid call.

STACK

**Backend** Python, FastAPI, async SQLAlchemy, Alembic, Pydantic, PostgreSQL, Redis + RQ, APScheduler, Playwright · **Frontend** React, TypeScript, Mantine, TanStack Query, Vite · **AI** OpenAI API, a shared MiniLM embedding service · **Infra** Docker Compose

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Built solo for an employer; partner and data details stay private.